

# Licensing of Code as Open Source in KTH-SAAB Collaborative Projects

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# Research Software Artifacts as Public Goods

- Many research processes produce artifacts as validation
- Software artifacts can be efficiently copied and distributed
- Software may become used by developers themselves
- Niche software often has unclear commercial potential
- Software patents are difficult to defend

# The Need for Clear Software Licenses for Use and Reuse

- Unclear licensing terms may paralyze users
- Engineers may be forbidden to use code with unclear license
- Clear licensing terms allow use and derived works
- Software license audits may happen at any time

# Free Software

Free software as defined by Free Software Foundation:

*“[U]sers have the freedom to run, copy, distribute, study, change and improve the software. [It] is a matter of liberty, not price.”*

<https://www.gnu.org/philosophy/free-sw.html>

# Open Source Software

Open source software as defined by Open Source Initiative:

- Free redistribution.
- Source code accessible.
- Derivative works allowed, possibly as patches.
- No discrimination between users.
- No discrimination between uses, including commercial use.
- Everyone with source is granted all open-source rights.
- License must cover all the code.
- No restrictions on other software distributed at the same time.
- Technological neutrality.

<https://opensource.org/osd>

# Permissive vs. Copyleft Licenses

**Permissive** Examples: BSD-3-Clause, MIT, Apache-2.0

**Copyleft** Derivative programs must use the same license.  
Examples: GPL-3.0-or-later

# Copyright Ownership of Software

A **copyright owner** of software can:

- License and distribute their software to others (for payment)
- Relicense their software on request
- Withhold new versions of software from others
- Transfer copyright ownership to others (including companies)

NB: If there are several owners, they must all agree on licensing and transfers.

# Licensee of Open Source Software

A **licensee** of open source software can:

- Distribute software to others under (at least) same license
- Sell the software to others (typically with same rights)
- Build derived works of software (usually under same license)



# Software Licensing at KTH

- Research software generally falls under Teacher's Exception
- Copyright is owned by the KTH employee(s)
- Employee may choose to license software as open source
- Copyright often held jointly by several employees
- Often no formal legal documentation of ownership

# Software Licensing at SAAB and in Industry

- Swedish law grants employer ownership of employee output
- Copyright is owned by SAAB unless legal agreement otherwise
- SAAB may choose to license software as open source
- Legal department may require documentation of ownership

# Scenarios and Issues

- Collaboratively developed software may be owned jointly by:
  - KTH employees
  - KTH students
  - SAAB
  - ...
- All may tacitly agree to license as open source
- But are ownership and licensing terms documented?
- What if there is a license audit?
- What if there is a commercial product using the software?

# Case Study: HOL4P4

- P4 is a network programming language
- HOL4P4 develops verification tools for P4
- Main developers:
  - Didrik Lundberg, SAAB (formerly KTH)
  - Anoud Alshnakat, PhD student, KTH
- Repository: <https://github.com/kth-step/hol4p4>
- License: Apache-2.0 or BSD-3-Clause

# Case Study: S3K

- S3K is an operating system kernel
- Main developer: Henrik Karlsson, PhD student, KTH
- Repository: <https://github.com/kth-step/s3k>
- License: MIT license

# Case Study: HolBA

- HolBA is a library for analysis of machine code (assembly)
- Main developers:
  - Didrik Lundberg, SAAB (formerly KTH)
  - Andreas Lindner, KTH
  - Karl Palmskog, KTH
- Repository: <https://github.com/kth-step/holba>
- License: BSD-3-Clause

# Possible Solutions and Models

- Software licensing terms agreed at project outset
- Explicit and legally valid documentation of copyright
- Process at SAAB for employee open source contributions?
- Framework for joint KTH-SAAB software maintenance?